Moderation Effects in the Association Between Prenatal WIC Participation and Child Mathematics Achievement

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Introduction

The Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) provides nutrition assistance to pregnant persons and young children from low-income families. Prior research often points to WIC participation as a predictor of better birth outcomes and early childhood development, although findings regarding long-term academic achievement are less consistent. This study investigates whether mothers' participation in WIC during pregnancy is related to children's mathematics achievement in 1997 and whether the relationship varies across important contextual factors.

Specifically, this report assesses three hypotheses of moderation:

1. whether the association between prenatal WIC participation and math achievement varies with family income.
2. whether it differs between White and Black children, and
3. whether scores change with the age of the child tested.

Understanding moderation is important, as an average effect may mask important subgroup differences: for example, WIC may benefit families with fewer resources, or program impacts may differ across demographic groups due to structural inequalities or differential access to complementary services. In addition, older children could show widening or narrowing gaps depending on cumulative environmental effects. By testing these interactions, this report provides a more nuanced view of how prenatal WIC exposure relates to later academic outcomes. This lab report uses the data from the Child Development Supplement of the Panel Study of Income Dynamics, focusing on the 1997 assessment year. Following instructions for this assignment, assumption diagnostics were conducted but not discussed in detail, with an

emphasis on data preparation, descriptive statistics, model estimation, and substantive interpretation.

Methods

Data and Variables

The analytic sample was constructed by selecting key variables required by the evaluation team: math achievement (mathraw97), WIC participation during pregnancy (WICpreg), family income (faminc97), race (CHRACE), age in 1997 (AGE97), birthweight indicator, and HOME environment scores. After restricting to complete cases, the final dataset contained N = 1,848 children.

We recoded race into a binary indicator (0 = White; 1 = Black) by following the given coding instruction. Because family income was highly skewed, we log-transformed and centered it. Similarly, child age and HOME scores were centered to facilitate the interpretation of interactions and to reduce multicollinearity. Continuous variables were checked for outliers and ranges representative of the underlying population.

Analytic Strategy

In order to answer the research questions, three different OLS regression models were estimated:

- **Model 1:** WIC × Income tested whether the relationship of WIC participation with math achievement depends on family income.
- **Model 2:** WIC × Race tested whether the relationship varies for Black versus White children.

- **Model 3:** WIC × Age tested whether the relationship changes as children age.

Each model controlled for the same set of covariates: centered age, centered income (as appropriate), race, birthweight, and HOME environment score. This ensures that any interaction effect reflects moderation rather than simple omitted variable bias. Results were summarized using stargazer to facilitate comparison across models. Although it is not a reporting requirement, model assumptions - linearity, homoscedasticity, normality, and influence - were investigated through standard diagnostic tools including residual plots and the Breusch-Pagan and White tests for heteroskedasticity. Minor deviations were addressed using centering transformations.

Results

Descriptive statistics

Descriptive statistics are presented below in Table 1 and suggest that there is considerable variation in this sample. Children's math scores ranged from 0 to 98 with a mean of 35.98 (SD = 22.34). Family income was highly right-skewed-mean ≈ $53,452; minimum = 0; maximum > $780,000-and thus warranted the log transformation used in the models. On average, the focal child was approximately 7.39 years old. HOME scores averaged 20.23, which is consistent with national norms.

Table 1. *Continuous Variables*

	mathraw97	AGE97	faminc97	HOME97
N	1848.00000	1848.000000	1848.00	1848.000000
Mean	35.97511	7.390152	53452.15	20.234903
SD	22.33576	2.921802	54909.07	3.096661
Min	0.00000	3.000000	0.00	7.900000
Max	98.00000	13.000000	784610.59	27.000000

Table 2 shows about 40% of mothers participated in WIC during pregnancy. The race composition was almost even: 52.5% White and 47.5% Black. As one might expect, the

birthweight and WIC participation vary enough so that interaction effects can be estimated without problems from perfect multicollinearity.

Table 2. *Categorical variables: frequencies and proportions*

Variable	Category	Frequency	Proportion
<chr>	<dbl>	<int>	<dbl>
WICpreg	0	1103	0.5968615
WICpreg	1	745	0.4031385
RACE	0	970	0.5248918
RACE	1	878	0.4751082
bthwht	0	1169	0.6325758
bthwht	1	679	0.3674242

Correlation patterns agree with theoretical expectations: both age and HOME scores show a positive correlation with math achievement; income is also positively related, albeit at a low magnitude. The correlation of math achievement and age, $r \approx 0.92$, reflects a strong developmental trend in cognitive scores during childhood.

Table 3. *Correlation Matrix*

	mathraw97	AGE97	faminc97	HOME97
mathraw97	1.0000000	0.91913234	0.15694496	0.3106784
AGE97	0.9191323	1.00000000	0.05260234	0.1920992
faminc97	0.1569450	0.05260234	1.00000000	0.3958002
HOME97	0.3106784	0.19209918	0.39580016	1.0000000

Regression Models

To better understand whether the effect of prenatal WIC participation differs across family and child characteristics, three moderation models were estimated and are summarized in Table 4. Each model includes the same core covariates, allowing a clear comparison of how income, race, and age shape the WIC - achievement relationship. The following subsections describe the results of each interaction and highlight how the association between WIC participation and math scores varies across groups.

Table 4. *Regression Models*

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	Dependent variable:		
	Math Achievement (mathraw97)		
	WIC × Income	WIC × Race	WIC × Age
	(1)	(2)	(3)
WICpreg	-1.421***	-2.589***	-1.306***
	(0.467)	(0.704)	(0.464)
faminc97_c	1.038***	0.553***	0.578***
	(0.260)	(0.185)	(0.185)
AGE97_c	6.955***	6.951***	7.129***
	(0.067)	(0.067)	(0.086)
RACE	-2.256***	-3.188***	-2.321***
	(0.462)	(0.574)	(0.460)
bthwht	-1.930***	-1.928***	-1.978***
	(0.403)	(0.404)	(0.403)
HOME97_c	0.597***	0.594***	0.611***
	(0.074)	(0.074)	(0.074)
WICpreg:faminc97_c	-0.912***		
	(0.348)		
WICpreg:RACE		2.178**	
		(0.888)	
WICpreg:AGE97_c			-0.440***
			(0.130)
Constant	38.102***	38.563***	38.275***
	(0.335)	(0.338)	(0.323)
Observations	1,848	1,848	1,848
R2	0.872	0.872	0.872
Adjusted R2	0.871	0.871	0.872
Residual Std. Error (df = 1840)	8.014	8.015	8.004
F Statistic (df = 7; 1840)	1,786.942***	1,785.989***	1,791.953***
Note:		*p<0.1; **p<0.05; ***p<0.01	

Model 1: Moderation by Family Income

In the WIC × Income model, WIC participation during pregnancy was significantly associated with lower math achievement ($\beta = -1.42$, $p = .002$). Family income was positively associated with math scores ($\beta = 1.04$, $p < .001$).

The interaction term (WIC × income) was significant and negative ($\beta = -0.91$, $p = .009$), indicating that the negative difference in math scores between WIC and non-WIC children is larger among higher-income families.

That suggests that:

- At low levels of income, WIC and non-WIC children score more similarly.
- At higher levels of income, children whose mothers did not use WIC score increasingly higher compared to those who did.

One interpretation is that WIC participation may serve as a marker of deeper unobserved disadvantage among higher-income households who would otherwise be expected to perform very well. Alternatively, WIC may provide more compensatory benefit for low-income families, reducing the performance gap relative to non-WIC peers.

Model 2: Moderation by Race

The second model showed strong main effects for WIC ($\beta = -2.59$, $p < .001$) and race ($\beta = -3.19$, $p < .001$), with Black children scoring lower on average. The interaction of WIC × race was statistically significant and positive, at $\beta = 2.18$ ($p = .014$).

This suggests that among Black children, the negative association of WIC participation with math achievement is weaker than it is among White children.

This means, substantively:

- For White children, the difference between WIC and non-WIC is relatively large and negative.
- That difference is smaller for Black children, which suggests the presence of a buffering or compensatory effect.

One possible explanation is that structural barriers and environmental constraints might affect academic outcomes disproportionately for Black families, so WIC participation provides relatively more support or reduces disparities for Black children compared to their White peers. This does not necessarily imply WIC eliminates racial disparities; rather, the WIC-associated gap in math scores is less pronounced among Black children.

Model 3: Moderation by Age

Child age was a strong predictor in all models, and particularly so here, $\beta = 7.13$, $p < .001$. The interaction, WIC × age, was significant and negative, $\beta = -0.44$, $p < .001$.

This suggests that the negative association between WIC participation and math scores becomes larger for older children. For younger children, say at age 5-6 years, the WIC versus non-WIC difference is small. At ages 10-13, it becomes more pronounced.

Several mechanisms may explain this pattern:

- Early developmental disadvantages may compound over time.
- WIC families may also continue to experience socioeconomic disadvantage that affects academic achievement over time.
- Differences in access to enrichment, schooling quality, and resources may widen as children age.

In spite of this widening gap, the overall effect of WIC remains modest, and its interaction effect-though statistically significant-is not overwhelmingly large in practical terms.

Model Comparison

All three models show excellent model fit, with an $R^2 \approx 0.872$ across specifications. Much of the explained variance is accounted for by age and HOME environment. The WIC effects and interaction terms are much smaller in magnitude than these environmental predictors, which suggests that developmental and home contexts bear importance for the shaping of academic outcomes.

Conclusion

This analysis suggests that the relationship between prenatal WIC participation and the children's math achievement varies meaningfully across family income, racial identity, and child age. While WIC participation is reliably associated with modestly lower math scores, these differences are not constant across groups. The WIC gap is larger among high-income families, smaller among Black children, and it grows with age.

These findings should not be interpreted as causal; unmeasured confounders such as maternal education, prenatal health, or neighborhood conditions likely affect both WIC participation and the children's academic performance. However, the results bring out important heterogeneity that program evaluators must take into consideration. Policies that evaluate the impacts of WIC or redesign nutritional support programs should avoid assuming a single homogeneous effect and instead recognize diversity in family contexts.

Future analyses might use quasi-experimental methods or longitudinal modeling to better isolate mechanisms of causation. Other moderators, such as maternal health behaviors, father

involvement, or early childhood education access, might further refine understanding of how prenatal nutritional support interacts with broader ecological factors.

Appendix

```
good <- read.csv("/Users/leeeileen/Desktop/Upenn/研一 fall/MSSP 8970/Assignment 3/good.csv")
regdata <- subset(
  good,
  select = c(mathraw97, WICpreg, CHRACE, AGE97, faminc97, bthwht, HOME97)
)
regdata$RACE <- ifelse(regdata$CHRACE == 9, NA,
                 ifelse(regdata$CHRACE == 1, 0,
                 ifelse(regdata$CHRACE == 2, 1, NA)))
regdata <- na.omit(regdata)
min_inc <- min(regdata$faminc97, na.rm = TRUE)
regdata$faminc97_log <- log(regdata$faminc97 - min_inc + 1)
regdata$AGE97_c      <- as.numeric(scale(regdata$AGE97,       center = TRUE, scale = FALSE))
regdata$faminc97_c   <- as.numeric(scale(regdata$faminc97_log,center = TRUE, scale = FALSE))
regdata$HOME97_c     <- as.numeric(scale(regdata$HOME97,      center = TRUE, scale = FALSE))
# N
nrow(regdata)
# Continuous variables: N, mean, sd, min, max
vars_cont <- c("mathraw97", "AGE97", "faminc97", "HOME97")
descriptives_cont <- sapply(
  regdata[vars_cont],
  function(x) c(
    N    = sum(!is.na(x)),
    Mean = mean(x),
    SD   = sd(x),
    Min  = min(x),
    Max  = max(x)
  )
)
descriptives_cont
# Categorical variables: frequencies and proportions
cat_table <- data.frame(
  Variable = c("WICpreg", "WICpreg",
               "RACE", "RACE",
               "bthwht", "bthwht"),

  Category = c(0, 1,
               0, 1,
               0, 1),

  Frequency = c(
    table(regdata$WICpreg)[1],
    table(regdata$WICpreg)[2],
    table(regdata$RACE)[1],
```

```
    table(regdata$RACE)[2],
    table(regdata$bthwht)[1],
    table(regdata$bthwht)[2]
  ),

  Proportion = c(
    prop.table(table(regdata$WICpreg))[1],
    prop.table(table(regdata$WICpreg))[2],
    prop.table(table(regdata$RACE))[1],
    prop.table(table(regdata$RACE))[2],
    prop.table(table(regdata$bthwht))[1],
    prop.table(table(regdata$bthwht))[2]
  )
)

cat_table

# Correlation matrix for continuous vars
cor_matrix <- cor(
  regdata[, c("mathraw97", "AGE97", "faminc97", "HOME97")],
  use = "pairwise.complete.obs"
)
cor_matrix
### 1. Moderation by family income
m_income <- lm(
  mathraw97 ~ WICpreg * faminc97_c + AGE97_c + RACE + bthwht + HOME97_c,
  data = regdata
)
summary(m_income)
### 2. Moderation by race
m_race <- lm(
  mathraw97 ~ WICpreg * RACE + AGE97_c + faminc97_c + bthwht + HOME97_c,
  data = regdata
)
summary(m_race)
### 3. Moderation by age
m_age <- lm(
  mathraw97 ~ WICpreg * AGE97_c + faminc97_c + RACE + bthwht + HOME97_c,
  data = regdata
)
summary(m_age)
stargazer(m_income, m_race, m_age, type = "text",
          dep.var.labels = "Math Achievement (mathraw97)",
          column.labels = c("WIC × Income", "WIC × Race", "WIC × Age"),
          digits = 3)
```